

Monster Game I

Time limit: 1.00 s

Memory limit: 512 MB

You are playing a game that consists of n levels. Each level has a monster. On levels $1, 2, \dots, n-1$, you can either kill or escape the monster. However, on level n you must kill the final monster to win the game.

Killing a monster takes sf time where s is the monster's strength and f is your skill factor (lower skill factor is better). After killing a monster, you get a new skill factor. What is the minimum total time in which you can win the game?

Input

The first input line has two integers n and x : the number of levels and your initial skill factor.

The second line has n integers $s_1, s_2, \dots, s_n$: each monster's strength.

The third line has n integers $f_1, f_2, \dots, f_n$: your new skill factor after killing a monster.

Output

Print one integer: the minimum total time to win the game.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq x \leq 10^6$
- $1 \leq s_1 \leq s_2 \leq \dots \leq s_n \leq 10^6$
- $x \geq f_1 \geq f_2 \geq \dots \geq f_n \geq 1$

Example

Input	Output
5 100 20 30 30 50 90 90 60 20 20 10	4800

Explanation: The best way to play is to kill the third and fifth monster.